

FIN Local Research

Checkpoint P428–P430

Type-Correct Cosine Reduction, an Exact-Rational
25-Variable Krawczyk Certificate, and Global Dual Feasibility

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*Local analytical, formal, and computer-assisted research.
No laboratory data, external audit, or physical validation.*

Contents

1	Executive summary	2
2	Epistemic convention	2
3	Inherited assumptions and updated state map	3
4	Reconstructed Legacy* and the coupling diagram	4
4.1	What the local source contains	4
4.2	Algebraic falsification inside the source	4
4.3	Correct present role of Legacy*	4
5	P428: a type-correct formal cosine boundary	5
5.1	Question, success, and falsification	5
5.2	Typing obstruction and repair	5
6	P429: exact-rational seven-contact Krawczyk box	6
6.1	The typed system	6
6.2	High-precision scout and rigorous enclosure	6
6.3	Certified data and theorem scope	7
7	P430: global contact-aware dual feasibility	7
7.1	Proof decomposition	7
7.2	Scoped optimality theorem	8
8	Newly constructed typed objects	8
8.1	O155: Rational-Cut Cosine Provider Interface	8
8.2	O156: Exact-Rational Seven-Contact KKT Isolating Box	9
8.3	O157: Contact-Aware Global Dual Certificate	9
9	Falsification attempts and failed approaches	10
9.1	Type attack	10
9.2	Conditioning attack	10
9.3	Exact-inverse expression swell	10
9.4	Global-grid attack	10
9.5	Legacy diagram attack	10
9.6	Alternative explanation	10
10	Numerical and formal reproducibility	10
11	Kernel, selector, dimensional, and operational audits	11

11.1 Kernel split	11
11.2 Selector	11
11.3 Dimensions	11
11.4 Dual dynamics and observer boundary	11
11.5 Physical evidence	11
12 Scientific interpretation after this checkpoint	12
13 Ranked next local programs	12
14 Artifact inventory	12
15 Restart instructions	13
16 Conclusion	13

1 Executive summary

This checkpoint completes Programs P428–P430 and converts the Release-10.35 simultaneous-contact candidate O144 from a floating-point construction into a rigorous, scoped computer-assisted result.

- **P428** – [**Proven reduction** / **Refuted interface** / **Locally blocked provider**]. Lean verifies all twelve FIN-specific rational phase conditions and reduces their cosine bounds to one generic alternating-series provider. The former interface `Rat → Rat` cannot denote ordinary real cosine at nonzero rational arguments. It is replaced by a rational-cut interface. The local Lean/Std installation has no real trigonometric analysis library, so the standard analytic provider is explicit but not proof-kernel compiled.
- **P429** – [**Computer-assisted proof**]. A 180-decimal Newton refinement followed by exact rational interval arithmetic proves a strict Krawczyk inclusion for all 25 variables of the seven-contact system. The first tested relative radius, 10^{-20} , succeeds. The weighted infinity contraction bound is $4.0673 \times 10^{-11} < 1$; all coordinates lie strictly inside the box; the six interior nodes are ordered; and the signed weights have pattern $(-, +, +, +, +, -, +)$. The certified objective box has width 4×10^{-20} around 0.7073534677231137.
- **P430** – [**Computer-assisted proof**]. Exact interval curvature treats six stationary contact neighborhoods, interval monotonicity treats the endpoint, and rational Bernstein ranges treat seven complementary intervals. Therefore the unique P429 contact polynomial satisfies

$$-1 \leq p(x) \leq 0 \quad (0 \leq x \leq 1).$$

Together with the inherited signed-moment weak-duality theorem, this proves global optimality of the certified primal–dual value in the declared moment problem. It does not prove that every global optimizer outside the isolating contact class is identical.

The batch also audits the reconstructed historical class

$$K_{\text{legacy}}^*(d) = A \frac{\cos(\pi d/4 + \pi/6)}{1 + \beta d}, \quad A, \beta > 0.$$

It is the research-backed reconstructed legacy class and a fixed-phase subclass of the binding intermediate $K_{\text{legacy,ont}}$. It is not the strict kernel. The local coupling diagram records four mechanisms and two parameter cross-modulations, but it defines no typed dynamical self-coupling map, feed-back equation, or fixed-point law. Its displayed path-exponent and integer-node calculations are algebraically false. The diagram is therefore historical intuition and provenance, not a self-coupling theorem or external physical evidence.

No complete legacy-to-strict bridge, physical-role transfer, non-premise selector, canonical orientation, dimensional unit, laboratory realization, total action, Standard Model, gravity theory, or Theory-of-Everything closure is obtained.

2 Epistemic convention

Status	Meaning in this checkpoint
[Proven]	Exact algebra, analytic theorem under explicit hypotheses, finite exhaustive result, or proof-kernel result.
[Computer-assisted proof]	A theorem whose finite enclosure conditions are checked by deterministic exact or directed interval computation.
[Strong evidence]	Stable bounded computation without a complete proof certificate.
[Conditional]	Valid only after a named source, state, reference, calibration, or axiom is supplied.

[Refuted]	The declared proposition fails in its stated class.
[Blocked locally]	A required formal library or typed provider is absent from the local environment.
[Blocked by external evidence]	Independent physical data, traceable calibration, custody, or unblinding is required.

Floating-point residuals are not theorems. Synthetic records validate software only. Historical statements in local prose are not admitted as external laboratory evidence.

3 Inherited assumptions and updated state map

The kernel split remains binding:

$$K_{\text{legacy,ont}}(d) = \alpha_{\text{geo}} \frac{\cos(\omega_L d + \phi_L)}{1 + \beta_{\text{tors}} d},$$

$$K_{\text{strict,gate}}(d) = \frac{\cos((743/4000)d + 13/80)}{1 + d^{9/5}}.$$

The reconstructed K_{legacy}^* is the fixed-phase legacy subclass

$$(\alpha_{\text{geo}}, \omega_L, \phi_L, \beta_{\text{tors}}) = (A, \pi/4, \pi/6, \beta).$$

The strict object remains the later completed/enriched working kernel. No silent equality or physical-role inheritance is used.

Lane	Before this batch	After this batch
Rational cosine arithmetic	Exact at twelve phases	Lean-checked and type-correctly reduced
Standard real cosine in Lean	Hidden behind Rat → Rat	Explicit rational-cut provider; local theorem absent
O144 contact system	Residual 3.42×10^{-28} , severely conditioned	Exact-rational Krawczyk existence and local uniqueness
O144 global dual	Sampling/plot only	Global interval Bernstein/curvature certificate
O144 objective	Floating midpoint	Enclosed to width 4×10^{-20}
Legacy-star provenance	Historical reconstruction and audits	Coupling and source audit integrated into current frontier
Legacy self-coupling	Intuitive language	No typed self-coupling law found
Selector / QW-2191	Open	Open
Dimensional physics	No internal unit source	Unchanged
Laboratory evidence	Absent	Absent

The ontology remains

$$\text{nadsoliton} \longrightarrow \text{light} \longrightarrow \text{matter} \longrightarrow \text{emergent observer},$$

with no lower informational layer below the nadsoliton.

4 Reconstructed Legacy* and the coupling diagram

4.1 What the local source contains

The historical source declares the multiplicative ansatz

$$K_{\text{total}} = K_{\text{geo}} K_{\text{res}} (1 + 0.2 K_{\text{tors}}) K_{\text{topo}}.$$

Factor	Diagram role	Source form	Audited status
K_{geo}	viscosity/damping	$e^{-\alpha d}$	historical label; no physical source theorem
K_{res}	resonance/phase syn-chronization	$1 + \alpha_{\text{res}} \text{sim}$	state/similarity law not typed
K_{tors}	oscillatory currents	$\cos(\pi d/4 + \pi/6)$	exact scalar carrier; no selector
K_{topo}	topological/path coupling	path sum, then $1/(1+\beta d)$	conditional reconstruction

The source also states

$$\alpha_{\text{res,eff}} = \alpha_{\text{res,base}} e^{-\alpha_{\text{geo}}/2}, \quad \beta_{\text{topo,eff}} = \beta_{\text{topo,base}} (1 + 0.5 |K_{\text{tors}}|).$$

These are genuine records of feedback or self-modulation intuition. They do not yet define mathematical self-coupling because the source supplies no state space X , self-map $S : X \rightarrow X$, time or iteration variable, update/flow equation, feedback closure condition, or fixed-point and stability theorem. The correct classification is **[Conditional]**: a cross-modulated factor ansatz, not self-coupled dynamics.

4.2 Algebraic falsification inside the source

The displayed exponents give

$$N(d) \sim d^{1.6}, \quad A_{\text{path}}(d) \sim d^{-0.6} \implies N(d) A_{\text{path}}(d) \sim d^{1.0},$$

which grows and cannot yield the d^{-1} tail of $1/(1+\beta d)$. Under the same path-count exponent, a d^{-1} total tail requires individual-path exponent -2.6 , not -0.6 .

Moreover,

$$\cos(\pi d/4 + \pi/6) = 0 \iff d = \frac{4}{3} + 4n.$$

The integer sequence 2, 5, 8, 11 asserted in the diagram is not a zero sequence. This refutes the displayed node calculation, not the use of a cosine carrier as an ansatz.

4.3 Correct present role of Legacy*

Under the historical role assignment and corrected path-sum reading, the admissible class is

$$\boxed{K_{\text{legacy}}^*(d) = A \frac{\cos(\pi d/4 + \pi/6)}{1 + \beta d}}, \quad A, \beta > 0.$$

The class is backed as a reconstruction of the intended legacy formula. Neither parameter is uniquely generated by the diagram. Values such as $A = 2.9$, $A = 4 \ln 2$, $\beta = 0.01$, $\beta = 0.05$, and $\beta = 1$ remain historical freezes, gauges, or comparisons unless a separate source theorem is supplied.

Legacy* exposes a linear hyperbolic envelope, fixed historical phase, and factor provenance. It does not source the strict phase (743/4000, 13/80), nonlinear exponent 9/5, target-independent positive damping scale, selector, physical units, or legacy physical-role transfer.

5 P428: a type-correct formal cosine boundary

5.1 Question, success, and falsification

Question. Can the twelve P411 rational Taylor intervals be connected to ordinary real cosine inside the dependency-free local Lean environment?

Success. A proof-kernel theorem, without a FIN-specific analytic axiom, that the twelve rational intervals enclose standard real cosine.

Falsification. A type mismatch, parity reversal, domain violation, or absence of the required real-analysis provider.

5.2 Typing obstruction and repair

The old interface `Rat -> Rat` cannot literally be ordinary cosine: $\cos q$ is generally irrational for nonzero rational q . P428 replaces it by

$$\text{CutValue} = (L, U, \text{compat}),$$

where $L(q)$ and $U(q)$ assert rational lower and upper bounds, with

$$L(\ell) \wedge U(u) \implies \ell \leq u.$$

The provider type is

$$\text{CosineCutProvider} : \mathbb{Q} \rightarrow \text{CutValue}.$$

For

$$x_d = \frac{743d + 650}{4000}, \quad d = 0, \dots, 11,$$

Lean checks exactly

$$0 \leq x_d, \quad x_d^2 < 12,$$

and for

$$S_n(x) = \sum_{k=0}^n (-1)^k \frac{x^{2k}}{(2k)!}$$

it proves

$$S_{21}(x_d) < S_{20}(x_d), \quad S_{20}(x_d) - S_{21}(x_d) < 10^{-30}.$$

The largest computed rational width is

$$1.9128718516949603 \times 10^{-37}.$$

The compiled theorem `twelve_bounds_from_standard_provider` reduces every FIN-specific bound to one generic provider on $0 \leq x, x^2 < 12$. The local Lean/Std stack contains no Mathlib or equivalent standard real trigonometric analysis library. Hence the status is:

- **[Proven]**: rational side conditions and dependency reduction;
- **[Refuted]**: `Rat -> Rat` as a type for ordinary cosine;
- **[Blocked locally]**: proof-kernel compilation of the standard real-cosine provider.

6 P429: exact-rational seven-contact Krawczyk box

6.1 The typed system

Let

$$z = (x_0, \dots, x_5, w_0, \dots, w_6, c_0, \dots, c_{11}) \in \mathbb{R}^{25}, \quad x_6 = 1,$$

and

$$p(x) = \sum_{j=0}^{11} c_j x^j.$$

The strict moment vector is

$$m_k = \frac{\cos(743k/4000 + 13/80)}{1 + k^{9/5}}, \quad k = 0, \dots, 11.$$

The equations are

$$\sum_{i=0}^6 w_i x_i^k = m_k, \quad k = 0, \dots, 11,$$

$$p(x_i) = t_i, \text{quad}(t_0, \dots, t_6) = (-1, 0, 0, 0, 0, -1, 0),$$

and

$$p'(x_i) = 0, \text{quadi} = 0, \dots, 5.$$

They define a continuously differentiable map $F : \mathbb{R}^{25} \rightarrow \mathbb{R}^{25}$.

6.2 High-precision scout and rigorous enclosure

At 180 decimal digits, Newton refinement gives successive maximum residuals

$$3.4204 \times 10^{-28}, \quad 6.0076 \times 10^{-73}, \quad 8.4790 \times 10^{-147}.$$

The double-precision condition estimate is about 6.65×10^{17} , so these residuals only locate the candidate.

Moments are enclosed by exact rational Taylor/Lagrange cosine bounds, exact rational fifth-root brackets for $k^{1/5}$, and exact interval propagation. The center z_0 is a 100-decimal rational vector and

$$r_i = 10^{-20} \max(1, |z_{0,i}|).$$

An exact inverse of the 100-decimal rational Jacobian was initially attempted, but caused severe expression swell on the available Intel i3-10110U/16 GB machine. The rigorous replacement uses a 180-digit numerical inverse rounded entrywise to a fixed 70-decimal rational matrix C . This is valid because a Krawczyk preconditioner may be arbitrary; all subsequent operations use the rounded C exactly.

For the rational box X , define

$$\mathcal{K}(z_0, X) = z_0 - CF(z_0) + (I - CDF(X))(X - z_0).$$

Exact interval arithmetic proves

$$\mathcal{K}(z_0, X) \subset \text{int } X.$$

The weighted infinity contraction estimate is

$$q = \max_i \sum_j \sup |\delta_{ij} - (CDF(X))_{ij}| \frac{r_j}{r_i} \leq 4.0672647472032875 \times 10^{-11} < 1.$$

The maximum image/box width ratio is bounded by the same value, and the minimum normalized inclusion margin is

$$0.9999999999593273.$$

6.3 Certified data and theorem scope

The nodes are enclosed around

$$\begin{aligned} &0.0198263786851421285, \quad 0.129422796249122243, \\ &0.295044257239125348, \quad 0.526692356352776609, \\ &0.814063214510439553, \quad 0.941844848017407955, \quad 1. \end{aligned}$$

The weights are enclosed around

$$(-0.626394136883480088, 0.619954368078206096, 0.592116744335684425, \\ 0.321132083364434028, 0.141956488001828879, -0.0809593308396336302, 0.0190196871332889238).$$

Their signs and strict node order are certified. The objective

$$\mathcal{N}_{\text{osc}} = -w_0 - w_5$$

lies in a rational interval of width 4×10^{-20} centered at

$$0.7073534677231137.$$

[**Computer-assisted proof**]: the exact moment/contact system has one and only one zero inside the declared 25-dimensional rational box. This is a local isolating theorem; global dual feasibility is a separate obligation.

7 P430: global contact-aware dual feasibility

7.1 Proof decomposition

A dense grid cannot exclude a narrow violation near an active contact. The proof splits $[0, 1]$ into rational contact neighborhoods of half-width 10^{-4} and their complement.

At x_0, x_5 , KKT equations give $p(x_i) = -1$. Interval Bernstein evaluation proves $p'' > 0$, hence these are convex minima and $p \geq -1$ locally; direct interval evaluation gives $p < 0$. At $x_1, \dots, x_4$, the equations give $p(x_i) = 0$ and interval curvature gives $p'' < 0$; these are concave maxima and $p \leq 0$, while direct evaluation gives $p > -1$.

Representative curvature enclosures are

$$1352.81 < p'' < 1367.85 \quad \text{near } x_0,$$

$$-103.59 < p'' < -102.64 \quad \text{near } x_1,$$

and

$$535.26 < p'' < 537.29 \quad \text{near } x_5.$$

At the endpoint, $p(1) = 0$ and on $[0.9999, 1]$

$$29.1457 < p'(x) < 29.1578.$$

Thus p increases toward zero and cannot cross above it; direct evaluation gives the lower bound.

On each complementary interval, exact affine power-to-Bernstein conversion and the Bernstein convex-hull property prove strict feasibility. All seven cells pass without recursive subdivision.

7.2 Scoped optimality theorem

Theorem (computer-assisted, inherited duality assumptions). Assume the standard real cosine enclosure theorem used by the moment constructor and the signed-measure weak-duality theorem in the P337–P380 chain. Then:

1. the 25-equation system has a unique zero in the P429 box;
2. its signed weights reproduce all twelve strict moments;
3. its polynomial satisfies $-1 \leq p \leq 0$ globally on $[0, 1]$;
4. primal and dual objectives agree by contact complementarity;
5. the enclosed value is globally optimal for the declared signed-moment problem.

The theorem does not assert global uniqueness of the optimizing measure outside the certified contact box. It has no direct physical interpretation.

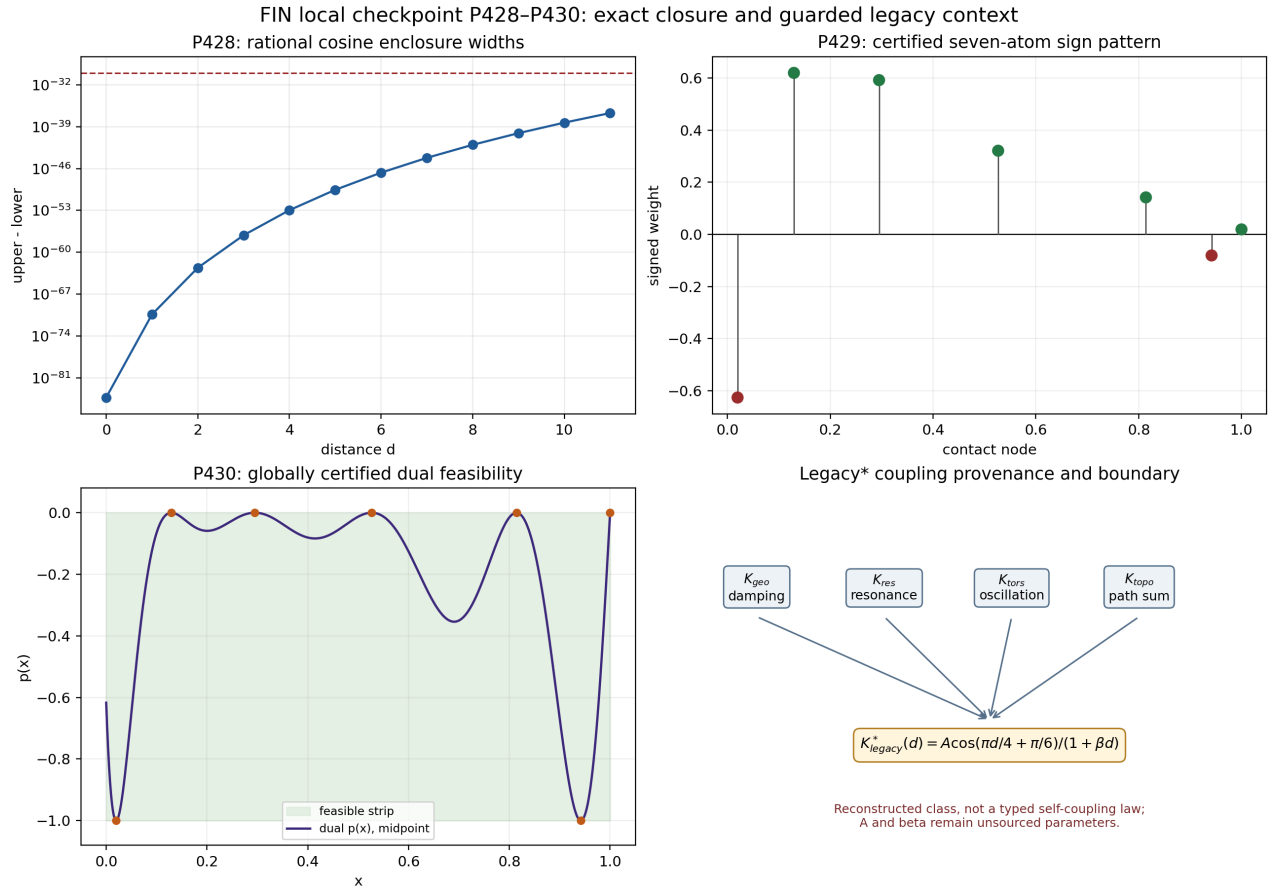


Figure 1: Exact closure of P428–P430 and the guarded Legacy* coupling context. The plotted polynomial is a midpoint visualization; the theorem comes from the interval cells, not from the curve.

8 Newly constructed typed objects

8.1 O155: Rational-Cut Cosine Provider Interface

Field	Definition/audit
Domain/codomain	$\mathbb{Q} \rightarrow \text{CutValue}$
Definition	Rational lower/upper predicates with compatibility
Required premise	Standard alternating-series cosine provider

Transformation law	Rational phase maps to its lower/upper cut
Generated theorem	All twelve P411 bounds from one provider
Necessity/removal	Reverting to Rat $\rightarrow$ Rat reintroduces the type error
Kernel relation	Strict moment phases only; no legacy substitution
Selector dependence	None; no orientation is selected
Dimensional status	Dimensionless
Operational meaning	Formal trust boundary, not apparatus
Failure mode	Provider absent or used outside its domain
Confidence	[Proven interface/reduction; provider locally blocked]

8.2 O156: Exact-Rational Seven-Contact KKT Isolating Box

Field	Definition/audit
Domain/codomain	Rational $X \subset \mathbb{R}^{25}$ and interval Krawczyk map
Definition	$\mathcal{K} = z_0 - CF(z_0) + (I - CDF(X))(X - z_0)$
Premises	Moment enclosures, differentiable KKT map, rational preconditioner
Transformation law	Requires explicit transformation of variables and weighted norm
Generated theorem	One unique KKT zero in X
Necessity/removal	Without inclusion/contraction, O144 remains numerical evidence
Kernel relation	Built from strict moments; Legacy* is contextual, not input
Selector dependence	None
Dimensional status	Dimensionless
Operational meaning	Exact optimization certificate, not measurement
Failure mode	Inclusion or contraction reaches the box boundary
Confidence	[Computer-assisted proof]

8.3 O157: Contact-Aware Global Dual Certificate

Field	Definition/audit
Domain/codomain	Interval coefficients/contact boxes to Boolean feasibility certificate
Definition	Curvature at contacts, monotonicity at 1, Bernstein hulls on complement
Premises	O156 and exact contact equations
Transformation law	Affine interval maps preserve Bernstein hull bounds
Generated theorem	$-1 \leq p \leq 0$ on $[0, 1]$
Necessity/removal	Without global cells, KKT stationarity is not dual feasibility
Kernel relation	Strict signed-moment dual only
Selector dependence	None
Dimensional status	Dimensionless
Operational meaning	Global mathematical witness, not observation
Failure mode	Any curvature, endpoint, or complement cell fails
Confidence	[Computer-assisted proof]

No self-coupling object is named for the legacy diagram because its state map and feedback law are absent.

9 Falsification attempts and failed approaches

9.1 Type attack

The P411 `Rat` $\rightarrow$ `Rat` abstraction was tested against ordinary cosine and rejected. The replacement represents irrational values by rational cuts without pretending the values are rational.

9.2 Conditioning attack

The KKT Jacobian is extremely ill-conditioned. Precision was escalated to 180 decimals, but residuals alone were explicitly rejected. Exact interval inclusion and the $q < 1$ contraction bound survived.

9.3 Exact-inverse expression swell

Direct inversion of the 100-decimal rational Jacobian produced excessive fraction growth on the declared hardware. The run was terminated without a scientific result. A fixed rounded rational preconditioner gives the same Krawczyk theorem interface and completed in seconds; no floating error is silently imported because the rounded matrix itself is used exactly.

9.4 Global-grid attack

A dense plot was not accepted as evidence. Direct wide-interval evaluation can overestimate near active contacts, so P430 separates contact geometry from noncontact Bernstein cells. All proof cells pass at depth zero.

9.5 Legacy diagram attack

The diagram was treated as a falsifiable source, not an authority. Its path exponent and integer-node sequence fail exact algebra. Its cross-modulations survive only as an ansatz-level self-coupling intuition.

9.6 Alternative explanation

The P429/P430 object is fully explained as an extremal solution of a finite signed-moment problem. Its existence does not establish a particle, field, thermodynamic entropy law, observer mechanism, or laboratory channel.

10 Numerical and formal reproducibility

Repeated complete runs produced byte-identical JSON and CSV outputs.

Component	Method
P428 side conditions	Exact Lean rational arithmetic and <code>native_decide</code>
P428 reflection	Python <code>Fraction</code> alternating sums

P429 root location	180-decimal <code>mpmath</code> Newton iteration
P429 moments	Exact fractions, cosine Taylor/Lagrange enclosure, exact fifth-root brackets
P429 preconditioner	70-decimal rational rounding of 180-digit inverse
P429 inclusion	Exact <code>Fraction</code> interval Krawczyk map
P429 uniqueness	Exact weighted infinity contraction $q < 1$
P430 contacts	Exact rational interval curvature/value bounds
P430 endpoint	Exact rational interval derivative/value bounds
P430 complement	Exact affine power-to-Bernstein transform
Regression	Eight new tests plus inherited suites

The local formal boundary is explicit: no real-cosine formal library and no Mathlib are present. No dependency was downloaded and no Internet service was used.

11 Kernel, selector, dimensional, and operational audits

11.1 Kernel split

P428–P430 consume strict moments. Legacy* appears only in the provenance audit. Neither kernel is substituted for the other. The completion still lacks source-independent phase/frequency, nonlinear compression, amplitude/damping, selector, and unit data. Physical-role transfer remains downstream and blocked.

11.2 Selector

O155–O157 are orientation-neutral certificates. They do not distinguish the two signs of the $\mathbb{Z}_{12}$ orientation torsor. QW-2191 remains open. The inversion-even radial Legacy* kernel also cannot supply a non-premise directed selector.

11.3 Dimensions

All phases, moments, nodes, weights, polynomial coefficients, and evolution parameters here are dimensionless. No action, length, time, mass, or energy unit is internally generated. A numerical scale is not a physical unit.

11.4 Dual dynamics and observer boundary

For a self-adjoint A ,

$$U_t = e^{-itA}, \quad P_t = e^{-tA}$$

are two functions of one spectral measure. This explains mathematical wave/heat duality but supplies no physical state, preparation, calibrated clock, reference frame, environment, instrument, apparatus response, measurement record, custody, or hold-out unblinding.

11.5 Physical evidence

No data in this checkpoint are laboratory data. References in the historical diagram to measurements, Wilson loops, or accuracy percentages are internal repository claims and are not admitted as physical evidence.

12 Scientific interpretation after this checkpoint

The batch materially strengthens the mathematical core:

1. the cosine trust interface is correctly typed;
2. the seven-contact system has certified local existence and uniqueness;
3. its dual is globally feasible;
4. the signed-moment optimum has a certified value enclosure;
5. Legacy* has explicit provenance and self-coupling boundaries.

It does not change the physical status of FIN. The deepest surviving interpretation is a finite spectral/operator and signed-moment framework with exact functional calculus, optimization certificates, and operational interfaces. An external typed extension by state, preparation, clock, reference frame, environment, instrument, apparatus response, dimensional standard, and independent record is still required before experimentally testable physics can be claimed.

13 Ranked next local programs

The next checkpoint should contain the bounded batch P435, P436, and P440. No external gate will be simulated as evidence.

Rank	Program	Decisive local output	Probability
1	P436	Rigorous certification of at least one positive O149 erasure-code gain cell	0.82
2	P440	Detector- and finite-sample-aware optimal JSR estimator, or minimax obstruction	0.70
3	P435	Genuine noisy-comb SDP primal/dual certificate on one nonideal cell, or local solver obstruction	0.62
4	P432	Exactly one complete-Bernstein/subordination damping-source class	0.58
5	P442	Representation-independence audit of O153 oriented record current	0.57
6	P433	Interval cover of one nontrivial photonic chart	0.54
7	P431	Formal Lindemann–Weierstrass provider; no suitable local library exists	0.10

The first action is a cheap dependency and invariant audit. An expensive SDP or minimax calculation begins only after its exact proposition and local solver path are known.

14 Artifact inventory

- FIN_Local_Research_Checkpoint_P428_P430_EN.md
- FIN_Local_Research_Checkpoint_P428_P430_EN.tex
- FIN_Local_Research_Checkpoint_P428_P430_EN.pdf
- FIN_Current_Local_Research_Report_EN.pdf
- RELEASE_10_36_PROGRAMS_428_430.md
- FIN_Programs_428_430_Results.json
- FIN_Programs_428_430_Summary.csv
- FIN_Programs_428_430_P428_Cosine.csv
- FIN_Programs_428_430_P429_Krawczyk.csv

- FIN_Programs_428_430_P430_Global_Dual.csv
- FIN_Programs_428_430_LegacyStar_Coupling_Audit.csv
- FIN_Programs_428_430_P428_Cosine_Reduction.lean
- FIN_Programs_428_430_Figures/p428_p430_exact_closure_and_legacy_context.png
- fin_programs_428_430.py
- test_fin_programs_428_430.py
- FIN_PROGRAMS_428_430_RELEASE_MANIFEST.sha256
- FIN_GOAL_STATE.md and the checkpoint guardrail in AGENTS.md

15 Restart instructions

From the repository root:

```
MPLCONFIGDIR=/tmp/fin-mpl-cache python3 fin_programs_428_430.py
.elan/toolchains/leanprover--lean4---v4.28.0/bin/lean \
  FIN_Programs_428_430_P428_Cosine_Reduction.lean
MPLCONFIGDIR=/tmp/fin-mpl-cache python3 -m unittest -v \
  test_fin_programs_428_430.py
lualatex -interaction=nonstopmode -halt-on-error \
  FIN_Local_Research_Checkpoint_P428_P430_EN.tex
lualatex -interaction=nonstopmode -halt-on-error \
  FIN_Local_Research_Checkpoint_P428_P430_EN.tex
sha256sum -c FIN_PROGRAMS_428_430_RELEASE_MANIFEST.sha256
```

Then inspect the state map and begin only the bounded P435/P436/P440 batch. Preserve the local-only rule, kernel split, QW-2191 boundary, dimensional boundary, and external-evidence gates.

16 Conclusion

O144 survives its decisive falsification tests. It is no longer merely a floating-point pattern: inside a declared rational box it has exact existence, local uniqueness, certified signs and ordering, and a globally feasible dual polynomial. The resulting value is a rigorous computer-assisted theorem for the declared dimensionless signed-moment problem.

The same rigor rejects overinterpretation. The reconstructed Legacy* kernel is a legitimate intermediate historical mathematical class, but its diagram provides neither typed self-coupling dynamics nor a source theorem for its free parameters. The strict/legacy bridge, selector, dimensions, operational apparatus, and empirical physics remain separate unsolved interfaces.